

inventory, only consumptive use by irrigated crops has been considered. Consumptive use in nonirrigated areas probably approximates the annual precipitation and, therefore, would not enter the recharge-discharge equation. The effective precipitation (table 18) was computed from the daily precipitation records of the U.S. Weather Bureau station at Madera by a method suggested by Blaney (1928, p. 154). In this method, 0.5 inch is deducted from the total precipitation for each storm, to account for evaporation. The remainder is considered effective for plant use.

TABLE 18.—*Unit consumptive-use values (feet) used in the inventory of recharge and discharge, Madera Canal service area, 1943, 1945, and 1952*

[Effective precipitation at Madera, in feet: 1943, 0.5; 1945, 0.4; 1952, 0.5. Estimated by deducting 0.5 in. for evaporation from total precipitation for each storm, as reported for Madera, Calif., by the U.S. Weather Bureau]

Crop	Unit consumptive use ¹ (including precipitation)	Unit consumptive use (less effective precipitation)		
		1943	1945	1952
Alfalfa.....	3.7	3.2	3.3	3.2
Irrigated pasture.....	3.7	3.2	3.3	3.2
Deciduous orchard.....	2.8	2.3	2.4	2.3
Vineyard.....	2.9	2.4	2.5	2.4
Cotton.....	2.5	2.0	2.1	2.0
Truck garden.....	1.8	1.3	1.4	1.3
Grain.....	1.0	.5	.6	.5
Milo.....	1.3	.8	.9	.8
Miscellaneous field crops.....	2.0	1.5	1.6	1.5

¹ Data from Bull. 2, California State Water Resources Board (1955, table 113).

TABLE 19.—*Estimated consumptive use of applied water by irrigated crops in the Madera Canal service area, 1943, 1945, and 1952*

Crop	1943			1945			1952		
	Unit consumptive use (feet)	Area irrigated (acres)	Consumptive use (acre-feet)	Unit consumptive use (feet)	Area irrigated (acres)	Consumptive use (acre-feet)	Unit consumptive use (feet)	Area irrigated (acres)	Consumptive use (acre-feet)
Alfalfa.....	3.2	22,800	73,000	3.3	24,800	81,800	3.2	23,300	74,600
Irrigated pasture.....	3.2	10,200	32,600	3.3	11,800	38,900	3.2	20,000	64,000
Deciduous orchard.....	2.3	3,000	6,900	2.4	4,200	10,000	2.4	21,600	51,800
Vineyard.....	2.4	19,800	47,500	2.5	21,300	53,200	2.0	62,700	125,400
Cotton.....	2.0	21,700	43,400	2.1	23,800	50,000	1.3	1,700	2,200
Truck garden.....	1.3	1,500	2,000	1.4	1,500	2,100	.5	2,300	1,200
Grain.....	.5	3,200	1,600	.6	(1)	-----	.8	600	400
Milo.....	.8	2,200	1,800	.9	(1)	-----	1.5	2,900	4,400
Miscellaneous field crops.....	1.5	5,200	7,800	1.6	6,800	10,900			
Total or weighted average.....	2.4	89,600	216,600	2.5	94,200	246,900	2.4	135,000	324,000

¹ None reported.

Infiltration was computed by deducting the consumptive use, which had been adjusted for effective precipitation, from the total water